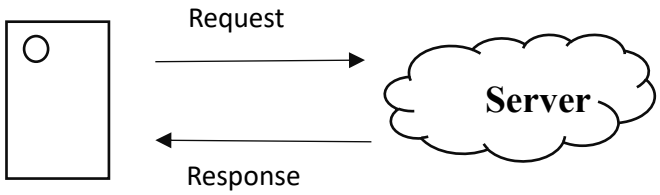
API Gateway

API : Application Programming Interface

API is a set of rules that allow two aaplications to communicate with each other.

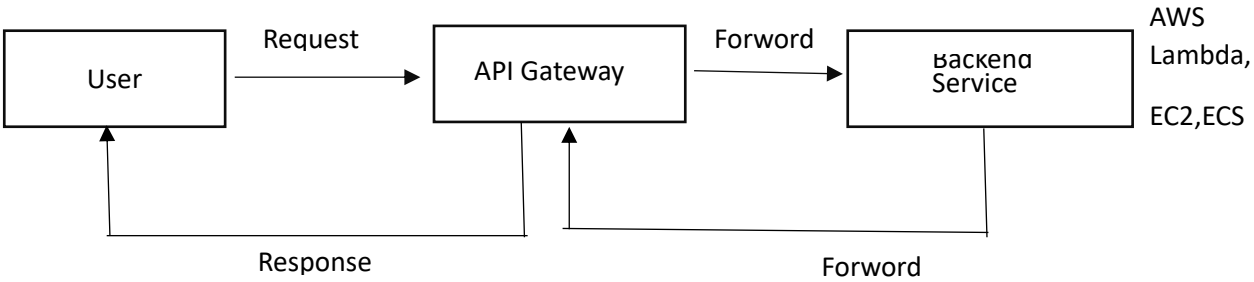
Example : We use mobile app to check weather.



In Short : Messenger between client and server.

API Gateway : API Gateway is a fully managed service used to create, publish,secure,monitor API's at any scale.

It act as a entry point for backend services like AWS Lambda,EC2,ECS and other HTTP/HTTPS Services



History :

Before API : Apps were standalone,could not share information.

API Came : Which helps apps to talk with each other.

Problem : When many apps talking with each other it becames confusing.

Solving : In 2015 Launched by AWS.

Today : API Gateway are cloud native ,secure and supporting REST API ,

WEBSOCKET .

Why we use?

We use API Gateway to control, secure,and manage requests before they reach backend service.

Types of API Gateway

1) REST API

Handles HTTP / HTTPS Requests
Most common type

Methods :

1) GET : Fetch Data
2) POST : Create Data
3) PUT : Update Data
4) DELETE : Remove Data
5) PATCH : Partial Data

2) WEBSOCKET

Supports real-time two way communication .

Example : Chat Application , Live Notification.

Implimentation :

Using HTTP :

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us-east-1.console.aws.amazon.com/apigateway/main/precreate?experience=rest&region=us-east-1

aws | Search [Alt+S] | United States (N. Virginia) | Account ID: 7601-5650-5473 | Yadav%20Arati

API Gateway > APIs > Create API

Choose an API type Info

HTTP API

Build low-latency and cost-effective REST APIs with built-in features such as OIDC and OAuth2, and native CORS support.

Works with the following:
Lambda, HTTP backends

Import | Build

WebSocket API

Build a WebSocket API using persistent connections for real-time use cases such as chat applications or dashboards.

Works with the following:
Lambda, HTTP, AWS Services

Build

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us-east-1.console.aws.amazon.com/apigateway/main/create?experience=rest&integration=usk7yw7&region=us-east-1&routes=xgr7zof&stage=$default

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API Gateway > APIs > Create API > Create HTTP API

Step 2 - *optional*
Configure routes

Step 3 - *optional*
Define stages

Step 4
Review and create

API details

API name
An HTTP API must have a name. The name is a non-unique value you use to identify and organize your APIs. To programmatically refer to this API, use the API ID that API Gateway generates for you.

API

IP address type | Info
Select the type of IP addresses that can invoke the default endpoint for your API. You don't need to redeploy your API for the update to take effect.

- IPv4
  Includes only IPv4 addresses.
- Dualstack
  Includes IPv4 and IPv6 addresses.

Integrations (1) Info

Specify the backend services that your API will communicate with. These are called integrations. For a Lambda integration, API Gateway invokes the Lambda function and responds with the response from the function. For an HTTP integration, API Gateway sends the request to the URL that you specify and returns the response from the URL.

HTTP | Remove

Method: ANY

URL endpoint: https://example.com

Add integration

Cancel | Review and create | Next

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API Gateway > APIs > Create API > Create HTTP API

Step 1
Configure API

Step 2 - *optional*
Configure routes

Step 3 - *optional*
Define stages

Step 4
Review and create

Configure routes - *optional*

Configure routes Info

API Gateway uses routes to expose integrations to consumers of your API. Routes for HTTP APIs consist of two parts: an HTTP method and a resource path (e.g., GET /pets). You can define specific HTTP methods for your integration (GET, POST, PUT, PATCH, HEAD, OPTIONS, and DELETE) or use the ANY method to match all methods that you haven't defined on a given resource.

Method	Resource path	Integration target	
ANY	/example	ANY https://example.com	Remove

Add route

Cancel | Review and create | Previous | Next

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us-east-1.console.aws.amazon.com/apigateway/main/create?experience=rest&integration=usk7yw7&region=us-east-1&routes=xgr7zof&stage=$default

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API Gateway > APIs > Create API > Create HTTP API

Step 1
Configure API

Step 2 - *optional*
Configure routes

Step 3 - *optional*
Define stages

Step 4
Review and create

Define stages - *optional*

Configure stages Info

Stages are independently configurable environments that your API can be deployed to. You must deploy to a stage for API configuration changes to take effect, unless that stage is configured to autodeploy. By default, all HTTP APIs created through the console have a default stage named $default. All changes that you make to your API are autodeployed to that stage. You can add stages that represent environments such as development or production.

Stage name	Auto-deploy	
$default	ON	Remove

Add stage

Cancel | Previous | Next

Ctrl V to paste

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API Gateway > APIs > Create API > Create HTTP API

Step 1
Configure API

Step 2 - *optional*
Configure routes

Step 3 - *optional*
Define stages

Step 4
Review and create

Review and create

API name and integrations

Edit

API name
API

IP address type
IPv4

Integrations
- ANY https://example.com (HTTP)

Routes

Edit

Routes

- ANY /example → ANY https://example.com (HTTP)

Stages

Stages

- $default (Auto-deploy: enabled)

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API Gateway Implementation S | API Gateway - Routes | Your Fortunecloud Training Tec

us-east-1.console.aws.amazon.com/apigateway/main/develop/routes?api=e463q16f7c&experience=rest&integration=usk7yw7&region=us-east-1&routes=xgr7zof&stage=$default

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API Gateway > APIs > Routes - API (e463q16f7c)

API Gateway

- APIs
- Custom domain names
- Domain name access associations
- VPC links

API: API(e463q16f7c)

- **Develop**
  - Routes
  - Authorization
  - Integrations
  - CORS
  - Reimport
  - Export
- **Deploy**
  - Stages
- **Monitor**
  - Metrics

Successfully created API API (e463q16f7c).

Routes

Stage: - | Deploy

Routes for API

Create

Search

▼ /example
ANY

Choose a route.

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us-east-1.console.aws.amazon.com/apigateway/main/apis?api=unselected&experience=rest&integration=usk7yw7&region=us-east-1&routes=odl7wwo&stage=$default

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API Gateway > APIs

API Gateway

- APIs
- Custom domain names
- Domain name access associations
- VPC links
- AgentCore targets New

- Usage plans
- API keys
- Client certificates
- Settings

Developer portals New

- Portals
- Portal products

Successfully created API API (e463q16f7c).

APIs (1/1)

Delete | Create API

Find APIs

Name	Description	ID	Protocol	API endpoint type	Created	Security policy	API status
API		e463q16f7c	HTTP	Regional	2026-01-11	-	-

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API Gateway Implementation S | API Gateway - Stage details | e463q16f7c.execute-api.us-eas | Your Fortunecloud Training Tec

us-east-1.console.aws.amazon.com/apigateway/main/deploy/stages?api=e463q16f7c&experience=rest&integration=usk7yw7&region=us-east-1&routes=odl7wwo&stage=$default

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API Gateway > APIs > API (e463q16f7c) > Stages

API Gateway

- APIs
- Custom domain names
- Domain name access associations
- VPC links

API: API(e463q16f7c)

- **Develop**
  - Routes
  - Authorization
  - Integrations
  - CORS
  - Reimport
  - Export
- **Deploy**
  - Stages
- **Monitor**
  - Metrics

Successfully created API API (e463q16f7c).

Stages

Stage: $default | Deploy

Stages for API | Create

Search

$default

Stage details | Delete | Edit

Details

Name	Created	Last updated
$default	January 11, 2026 7:16 PM	January 11, 2026 7:16 PM

Invoke URL

https://e463q16f7c.execute-api.us-east-1.amazonaws.com

Description

None

Attached deployment

Automatic Deployment

Enabled

Deployment ID	Deployment created
alepit	January 11, 2026 7:16 PM

Deployment description

Automatic deployment triggered by changes to the Api configuration

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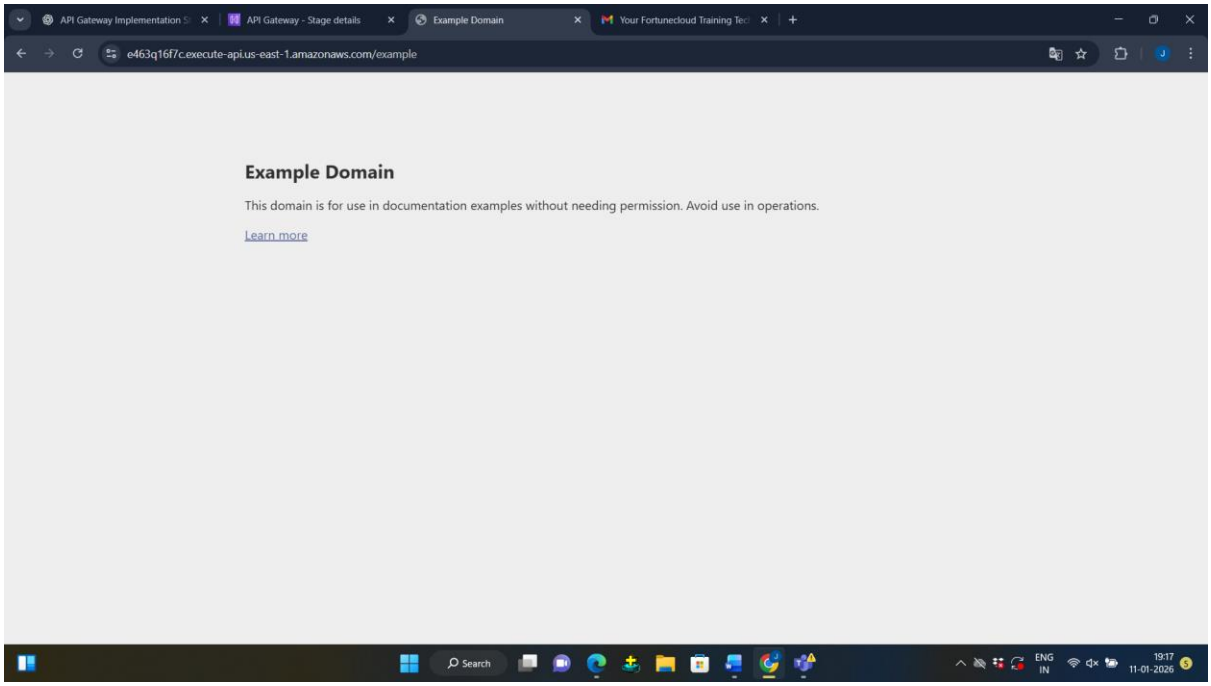
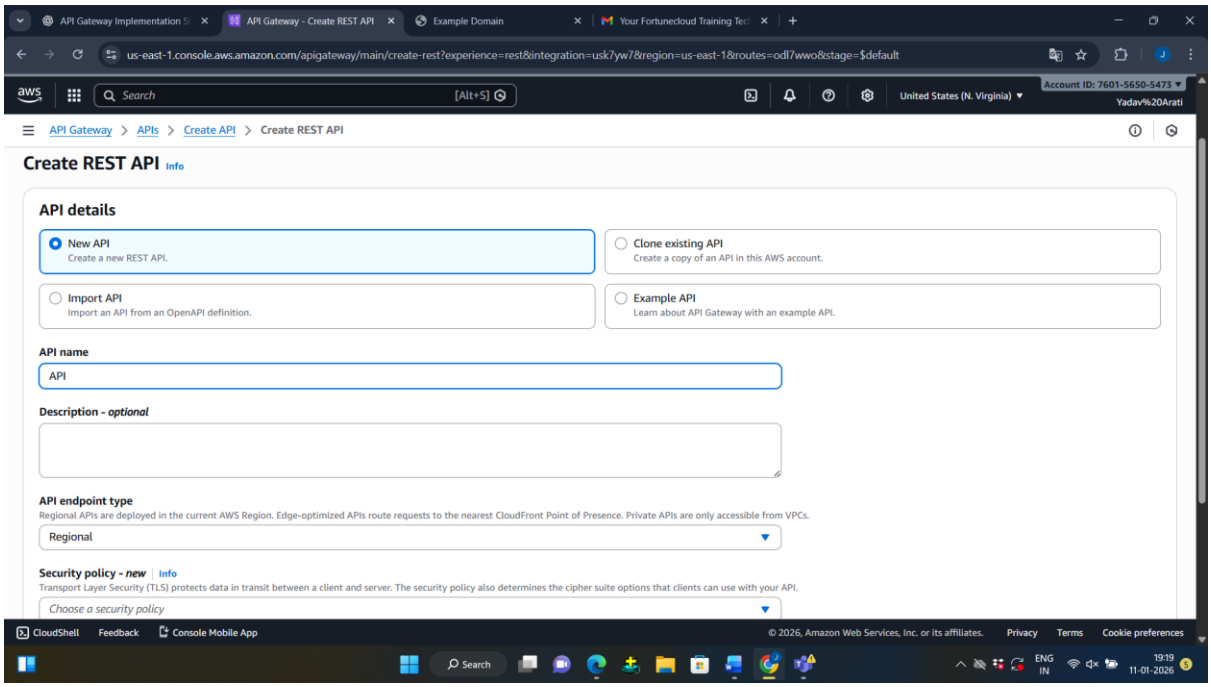


Using REST API :



API Gateway Implementation S | API Gateway - Create method | Example Domain | Your Fortunecloud Training Tec

us-east-1.console.aws.amazon.com/apigateway/main/apis/1ejvngxx9c/resources/pa71p32ph8/create-method?api=1ejvngxx9c&experience=rest&integration=usk7yw7&region=us-east-1...

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API Gateway > APIs > Resources - API (1ejvngxx9c) > Create method

Create method

Method details

Method type

ANY

Integration type

- Lambda function
  Integrate your API with a Lambda function.
- HTTP
  Integrate with an existing HTTP endpoint.
- Mock
  Generate a response based on API Gateway mappings and transformations.
- AWS service
  Integrate with an AWS Service.
- VPC link
  Integrate with a resource that isn't accessible over the public internet.

HTTP proxy integration
Send the request to your HTTP endpoint without customizing the integration request or integration response.

Response transfer mode | Info

Buffered

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API Gateway Implementation S | API Gateway - Create method | Example Domain | Your Fortunecloud Training Tec

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API Gateway > APIs > Resources - API (1ejvngxx9c) > Create method

HTTP method

ANY

Endpoint URL

https://example.com

Content handling | Learn more

Passthrough

Integration timeout | Info

By default, you can enter an integration timeout of 50 - 29,000 milliseconds. You can use Service Quotas to raise the integration timeout to greater than 29,000 ms

29000

- Method request settings
- URL query string parameters
- HTTP request headers
- Request body

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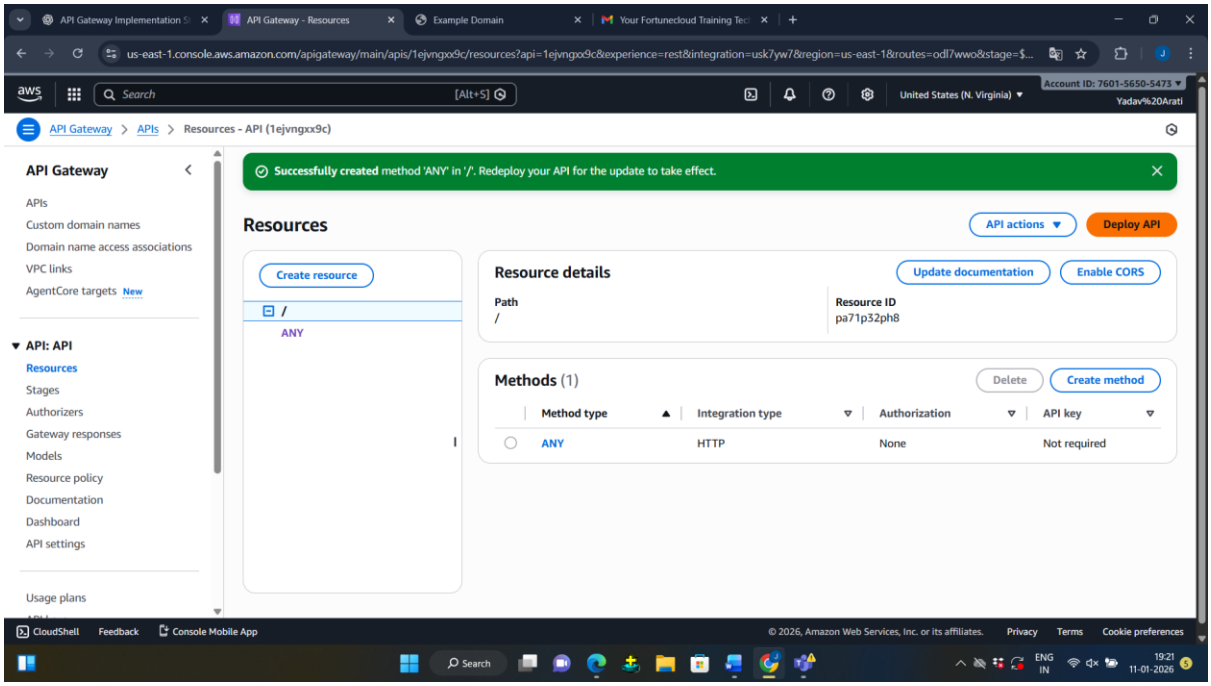
API Gateway Implementation S
API Gateway - Resources
Example Domain
Your Fortunecloud Training Tec
us-east-1.console.aws.amazon.com/apigateway/main/apis/1ejvngxx9c/resources?api=1ejvngxx9c&experience=rest&integration=usk7yw7&region=us-east-1&routes=odl7wwo&stage=$...
Search
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API Gateway > APIs > Resources - API (1ejvngxx9c)
API Gateway
APIs
Custom domain names
Domain name access associations
VPC links
AgentCore targets New
API: API
Resources
Stages
Authorizers
Gateway responses
Models
Resource policy
Documentation
Dashboard
API settings
Usage plans
Successfully created method 'ANY' in '/'. Redeploy your API for the update to take effect.
Resources
API actions
Deploy API
Create resource
/
ANY
Resource details
Update documentation
Enable CORS
Path
/
Resource ID
pa71p32ph8
Methods (1)
Delete
Create method
Method type
Integration type
Authorization
API key
ANY
HTTP
None
Not required
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19:21
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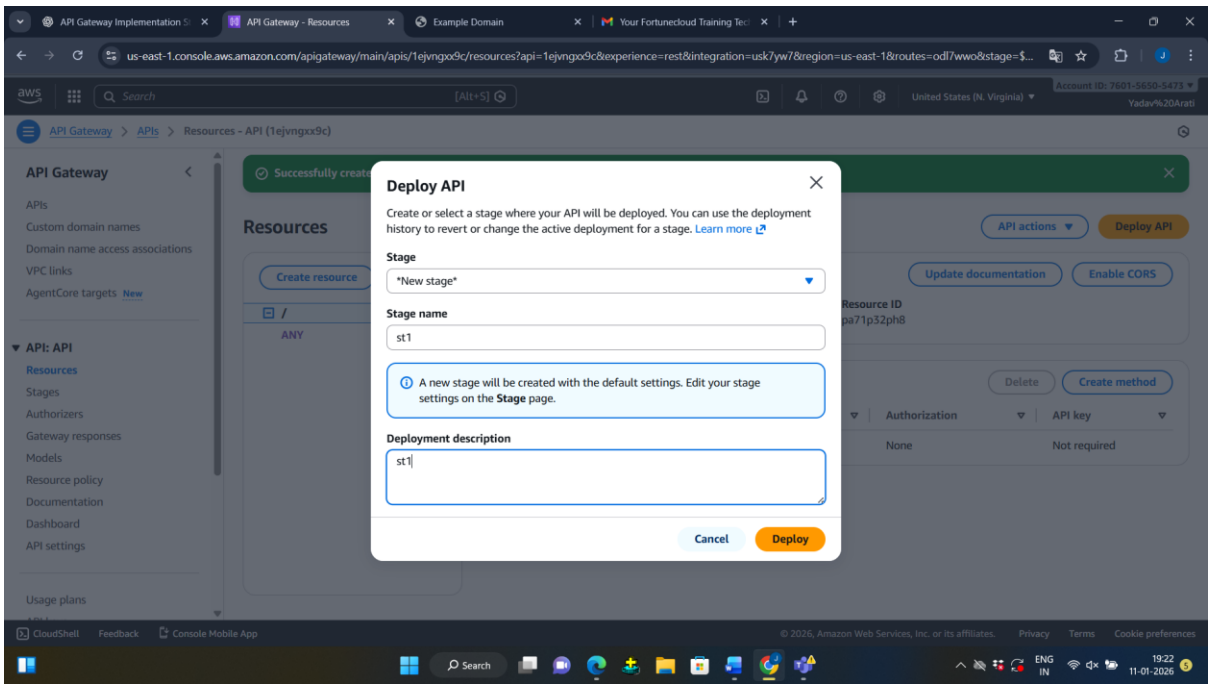
API Gateway Implementation S
API Gateway - Resources
Example Domain
Your Fortunecloud Training Tec
us-east-1.console.aws.amazon.com/apigateway/main/apis/1ejvngxx9c/resources?api=1ejvngxx9c&experience=rest&integration=usk7yw7&region=us-east-1&routes=odl7wwo&stage=$...
Search
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API Gateway > APIs > Resources - API (1ejvngxx9c)
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Documentation
Dashboard
API settings
Usage plans
Resources
Create resource
/
ANY
API actions
Deploy API
Update documentation
Enable CORS
Resource ID
pa71p32ph8
Delete
Create method
Authorization
API key
None
Not required
Deploy API
Create or select a stage where your API will be deployed. You can use the deployment history to revert or change the active deployment for a stage. Learn more
Stage
New stage
Stage name
st1
A new stage will be created with the default settings. Edit your stage settings on the Stage page.
Deployment description
st1
Cancel
Deploy
CloudShell
Feedback
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19:22
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API Gateway Implementation S | API Gateway - Stages | Example Domain | Your Fortunecloud Training Tec

us-east-1.console.aws.amazon.com/apigateway/main/apis/1ejvngxx9c/stages?api=1ejvngxx9c&experience=rest&integration=usk7yw7&region=us-east-1&routes=odl7wwo&stage=%24d...

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API Gateway > APIs > API (1ejvngxx9c) > Stages

API Gateway

- APIs
- Custom domain names
- Domain name access associations
- VPC links
- AgentCore targets New

API: API

- Resources
- Stages
- Authorizers
- Gateway responses
- Models
- Resource policy
- Documentation
- Dashboard
- API settings

Usage plans

Successfully created deployment for API. This deployment is active for st1.

0 0 2 0 0

Stages

Stage actions | Create stage

st1

Stage details Info

Edit

Stage name	Rate Info	Web ACL
st1	10000	-
Cache cluster Info	**Burst Info**	**Client certificate**
Inactive	5000	-
Default method-level caching		
Inactive		

Invoke URL
https://1ejvngxx9c.execute-api.us-east-1.amazonaws.com/st1

Active deployment
8x6f9b on January 11, 2026, 19:23 (UTC+05:30)

Logs and tracing Edit

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API Gateway Implementation S | API Gateway - Stages | Example Domain | Your Fortunecloud Training Tec

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Example Domain

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Learn more

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